

QCE Chemistry Units 3&4

Test 1: Chemical equilibrium systems

- Chemical equilibrium
- Factors that affect equilibrium
- Equilibrium constants
- Properties of acids and bases
- pH scale

Student's Name: _____

Teacher's Name: _____

General instructions

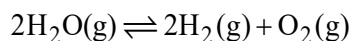
- Write your name and your teacher's name in the space provided above on this page.
 - A formula and data booklet is provided.
 - Unless otherwise indicated, the diagrams in this booklet are not drawn to scale.
 - Answer all questions in the spaces provided.
- Test time – 45 minutes
 - Total marks – 40 marks

SECTION 1**Instructions**

- Choose the best answer for Questions 1–10.
 - This section has 10 questions and is worth 10 marks.
 - Circle A, B, C or D.
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QUESTION 1

The equilibrium equation for a reaction is shown.



What is the equilibrium expression for this reaction?

- (A) $K_c = \frac{[\text{H}_2\text{O}]^2}{[\text{H}_2]^2 \times [\text{O}_2]}$
- (B) $K_c = \frac{[2\text{H}_2\text{O}]}{[\text{H}_2] \times [\text{O}_2]}$
- (C) $K_c = \frac{[2\text{H}_2] \times [\text{O}_2]}{[2\text{H}_2\text{O}]}$
- (D) $K_c = \frac{[\text{H}_2]^2 \times [\text{O}_2]}{[\text{H}_2\text{O}]^2}$

QUESTION 2

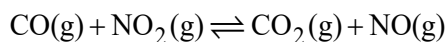
The equilibrium constant (K_c) of a reaction is 7.3×10^{-8} .

At equilibrium, the

- (A) reactants and products are present in equal amounts.
- (B) reactants are favoured.
- (C) rate of the forward reaction is greater than the rate of the reverse reaction.
- (D) products are favoured.

QUESTION 3

The equilibrium equation for a reaction is shown.

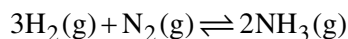


If the pressure is decreased, the equilibrium position will

- (A) shift to the left.
- (B) shift to the right.
- (C) remain the same.
- (D) no longer exist.

QUESTION 4

The reaction to produce ammonia gas (NH_3) at 270 K is represented by the following equation.



15 mol of H_2 is mixed with 15 mol of N_2 in a sealed 1.0 L container. At equilibrium, there are 12 mol of H_2 remaining.

What amounts of N_2 and NH_3 would be present at equilibrium?

	N_2	NH_3
(A)	12 mol	3 mol
(B)	3 mol	6 mol
(C)	14 mol	2 mol
(D)	9 mol	9 mol

QUESTION 5

Which of the following would change the equilibrium constant (K_c) of a reaction?

- (A) adding a catalyst
- (B) increasing pressure
- (C) increasing the volume of the container
- (D) increasing temperature

QUESTION 6

A solution with a pH of 3 will have

- (A) twice the hydrogen ion (H^+) concentration of a solution with a pH of 5.
- (B) four times the H^+ concentration of a solution with a pH of 5.
- (C) 10 times the concentration H^+ concentration of a solution with a pH of 5.
- (D) 100 times the H^+ concentration as a solution with a pH of 5.

QUESTION 7

Determine the H^+ concentration of a solution with a pH of 2.8.

- (A) $3.4 \times 10^{-7} \text{ M}$
- (B) $1.6 \times 10^{-3} \text{ M}$
- (C) 0.53 M
- (D) 1.0 M

QUESTION 8

Calculate the pOH of a solution with a pH of 6.

- (A) 4
- (B) 6
- (C) 8
- (D) 10

QUESTION 9

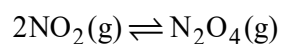
A chemical reaction occurs in a closed system and reaches equilibrium.

For this system, which of the following observations is least likely to be made?

- (A) constant colour
- (B) equal amounts of reactants and products
- (C) reactions still occurring
- (D) constant temperature

QUESTION 10

The reaction shown has an equilibrium constant (K_c) of 1000 at a given temperature.



Determine the K_c for the reaction $2\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 4\text{NO}_2(\text{g})$.

- (A) 1×10^{-6}
- (B) $\frac{1}{1000}$
- (C) 1000
- (D) 1×10^6

SECTION 2**Instructions**

- This section has nine questions and is worth 30 marks.

QUESTION 11 (5 marks)

The table shows the percentage of product present in an equilibrium mixture at different temperatures and pressures.

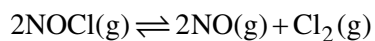
Pressure (atm)	Product present (%)			
	200°C	250°C	300°C	350°C
1	10	3.7	1.2	0.4
100	63	28	21	17
200	97	61	39	19

- a) State whether the forward reaction is endothermic or exothermic. Explain your reasoning. [3 marks]

- b) Identify and explain whether the reactants or products have more gaseous particles. [2 marks]

QUESTION 12 (4 marks)

A sealed vessel contains a system at equilibrium at 370 K. The reaction is shown below.



The concentrations of the gases at equilibrium are shown in the table.

	NOCl	NO	Cl ₂
Concentration (mol L ⁻¹)	0.067	2.65	3.22

The reaction is repeated in another sealed vessel at the same temperature with different concentrations of the gases. At equilibrium, the concentration of Cl₂ is 2.76 mol L⁻¹ and the concentration of NO is 1.79 mol L⁻¹.

Determine the concentration of NOCl in the second vessel. Show your working.

Concentration = _____ mol L⁻¹ (to three significant figures)

QUESTION 13 (5 marks)

At 30°C, the self-ionisation constant of water (K_w) is 1.47×10^{-14} . At 100°C, K_w is 51.3×10^{-14} .

- a) Calculate the pH of water at both temperatures. Show your working. [3 marks]

pH at 30°C = _____ (to three significant figures)

pH at 100°C = _____ (to three significant figures)

- b) Use your answer for 13a) to explain whether the ionisation of water is endothermic or exothermic. [2 marks]

QUESTION 14 (1 mark)

Explain why the pH of a weak acid solution cannot be determined using the concentration of the acid only.

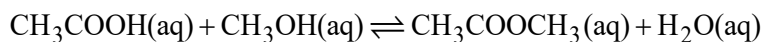
QUESTION 15 (2 marks)

When limestone (calcium carbonate) is heated, it decomposes to form calcium oxide and carbon dioxide. In an open container, complete decomposition of limestone occurs; however, in a sealed container, equilibrium is established.

Explain why this occurs.

QUESTION 16 (4 marks)

Methyl ethanoate ($\text{CH}_3\text{COOCH}_3$) is formed according to the reaction shown.



The equilibrium constant (K_c) of the reaction is 6.

In a series of experiments, different concentrations of the reactants were reacted. The results were recorded in the table.

Experiment	CH_3COOH (mol L^{-1})	CH_3OH (mol L^{-1})	$\text{CH}_3\text{COOCH}_3$ (mol L^{-1})	H_2O (mol L^{-1})
1	1	1	2	2
2	4	0.5	3	4
3	1.5	1	5	3
4	0.5	2	3	2

For each experiment, determine whether the reaction reached equilibrium. For any reaction that is not at equilibrium, state the direction that the reaction would need to proceed to reach equilibrium.

Show your working.

Experiment 1 _____

Experiment 2 _____

Experiment 3 _____

Experiment 4 _____

QUESTION 17 (3 marks)

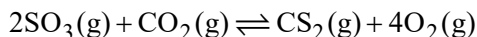
A sample of bromine chloride (BrCl) gas decomposes to form chlorine (Cl_2) gas and bromine (Br_2) gas in a 1 L container. At 300 K, the equilibrium constant (K_c) for the reaction is 13.7 and the equilibrium mixture contains 4.60 mol of Cl_2 .

Determine the amount of Br_2 and BrCl in the mixture at 300 K. Show your working.

amount of Br_2 = _____ mol (to three significant figures)
amount of BrCl = _____ mol (to three significant figures)

QUESTION 18 (4 marks)

Carbon disulfide (CS₂) can be produced using the endothermic reaction between carbon dioxide (CO₂) and sulfur trioxide (SO₃).



State the effect that the following changes would have on the equilibrium position once the reaction is at equilibrium.

- a) addition of SO₃ [1 mark]

- b) addition of a catalyst [1 mark]

- c) a decrease in volume [1 mark]

- d) an increase in temperature [1 mark]

QUESTION 19 (2 marks)

In an experiment, sulfuric acid (H₂SO₄(l)) and water (H₂O(l)) were found to be poor conductors of electricity. However, a mixture of H₂SO₄(aq) in H₂O(l) was found to be a good conductor of electricity.

Explain these findings.

END OF TEST